EXERCISE 1 - ANSWERS

1. SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jana	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name,
salary
FROM employees
ORDER BY salary DESC;

first_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4 SELECT *
FROM ~~EMPLOY~~ employees
ORDER BY salary DESC
LIMIT 5;

id	first_name	last_name	department	salary	hire_date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angele
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

5. SELECT *
FROM employees
WHERE department = 'IT';

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2021-03-25	New York
4	Sarah	Brown	IT	53000	2021-02-14	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	60000	2020-08-10	San Francisco

6. SELECT *
FROM employees
WHERE department = 'Finance' AND salary > 58000;

id	first_name	last_name	department	salary	hire_date	city
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

7. SELECT *
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

id	first_name	last_name	department	salary	hire_date	city
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

8. SELECT *
FROM employees
WHERE NOT department = 'IT';

id	first_name	last_name	department	salary	hire_date	city
2	Jane	Smith	HR	48000	2019-07-20	~~Los Angeles~~ Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Chicago Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	54000	2019-10-01	Houston
8.	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

9. SELECT*
FROM employees
WHERE department IN ('HR','IT','Finance');

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-07-20	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	54000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

10. SELECT* FROM employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'New York'

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

11. SELECT first_name,
last_name
FROM employees
WHERE (department = 'Finance' OR department = 'Marketing')
AND salary > 52000
ORDER BY salary DESC;

first_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12. SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');

City
Los Angeles
San Francisco
Houston
Chicago

13. SELECT *
FROM employees
WHERE department NOT 'Finance'
AND SALARY > 50000
ORDER BY hire_date ASC;

id	first-name	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
1	John	Doe	IT	65000	2018-06-15	New York
8	Jessica	Moore	Moore	51000	2019-05-22	Los Angeles
4	Sarah	Brown	~~Broo~~ IT	52000	2021-03-25	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

14 SELECT * FROM employees
WHERE city IN ('chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;

id	first-name	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago